



Virtual Internship RevoU x Telkom Indonesia
Project 2

Active User Behavior & Segmentation – OCA Indonesia

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This Project is part of the RevoU x Telkom Indonesia Virtual Internship Program

Business Background

What is OCA Indonesia?

OCA Indonesia is a Communication Platform-as-a-Service (CPaaS) owned by Telkom Indonesia. It enables businesses to send messages at scale through four main channels, powering notifications, marketing campaigns, and operational communications for enterprise clients.

Revenue Model

OCA generates revenue based on billable messages. Only successfully delivered messages are charged, making delivery rate a critical KPI directly tied to business performance.

WhatsApp

Marketing Service Utility

SMS

Batch blast High deliverability

Email

Delivered Opened Clicked

Call

IVR Answered / RNA

Dataset scope: 20 enterprise clients 4 channel transaction tables ~122,700 total transactions Period: January - March 2025

Business Problem & Analysis Objectives

BUSINESS PROBLEM

1. OCA needs to understand how clients differ in communication behavior to allocate resources effectively.
2. Not all clients generate equal revenue -- identifying high-value vs. at-risk clients is critical.
3. Without segmentation, product and sales teams lack a data-driven basis for personalization and upsell strategies.
4. A drop in delivery rate or client engagement signals revenue risk that must be detected early.

ANALYSIS OBJECTIVES

Segment clients -- using FM-based RFM to identify Power, Active, and Regular users

Identify profitable segments -- and quantify their contribution to total platform revenue

Detect churn risk -- by analyzing recency, volume trends, and delivery performance

Find upsell potential -- for clients underutilizing high-revenue channels like WhatsApp & SMS

Dataset Scope & Chosen Metrics

wa_transactions

50,000 rows 7 cols

message_type, price, last_status,
is_charge

sms_transactions

20,000 rows 7 cols

total_sms, total_price, message_status

email_transactions

40,000 rows 6 cols

price, message_status, is_charge

call_transactions

12,749 rows 6 cols

total_price, message_status
(answered/rna)

users

20 rows 4 cols

user_name, field_of_business, join_date

Why These Metrics?

Monetary (M) (*total_revenue*) -- Primary revenue driver -- most differentiating across clients

Delivery Rate (*charged/total tx*) -- Business health signal; low rate = revenue & trust risk

Frequency (F) (*total_transactions*) -- Proxy for usage intensity -- volume of blast activity

Channel Mix % (*revenue per channel*) -- Identifies dominant channel and upsell opportunity

Methodology

01

Data Extraction (BigQuery SQL)

Queried 4 separate channel tables (Whatsapp, SMS, Email, Call) and joined with the Users table. Computed per-user metrics: transaction counts, charged transactions, revenue per channel, active days, and last active date.

02

Feature Engineering

Derived channel revenue share (%), dominant channel by revenue, WA delivery rate, Call answer rate, and revenue per transaction -- creating a 20-row user profile table.

03

FM Scoring (Python)

Scored each user on Frequency (total_transactions) and Monetary (total_revenue) using 33rd/66th percentile thresholds. R score omitted -- all users equally recent (same last active date).

04

Rule-Based Segmentation

FM Total 6 Power User | FM Total 4 Active User | FM Total < 4 Regular User. Threshold justification: natural 3-tier gap in revenue data (Rp300K / Rp1.3M / Rp5.2M).

05

Visualization & Insight Generation

Generated scatter plot, revenue breakdown, segment summary chart, and hourly line chart. Insights derived and mapped to actionable business recommendations.

Root Cause Analysis

Why do OCA clients differ significantly in engagement & revenue contribution?

CLIENT ENGAGEMENT DISPARITY

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graph LR; A[CLIENT ENGAGEMENT DISPARITY] --- B[Usage Volume Gap]; A --- C[Channel Effectiveness]; A --- D[Behavioral Timing]; B --- B1[Power Users generate 16 more revenue than Regular Users]; B --- B2[Tx volume gap: 24,635 vs 1,500 per quarter]; C --- C1[Call shows 76% RNA rate -- high cost, low return]; C --- C2[SMS drives 46% of revenue but requires volume to be viable]; D --- D1[Power Users blast at 08-11 vs Regular at 12-16]; D --- D2[~79% of all blasts within business hours across segments];
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Usage Volume Gap

Power Users generate 16 more revenue than Regular Users

Tx volume gap: 24,635 vs 1,500 per quarter

Channel Effectiveness

Call shows 76% RNA rate -- high cost, low return

SMS drives 46% of revenue but requires volume to be viable

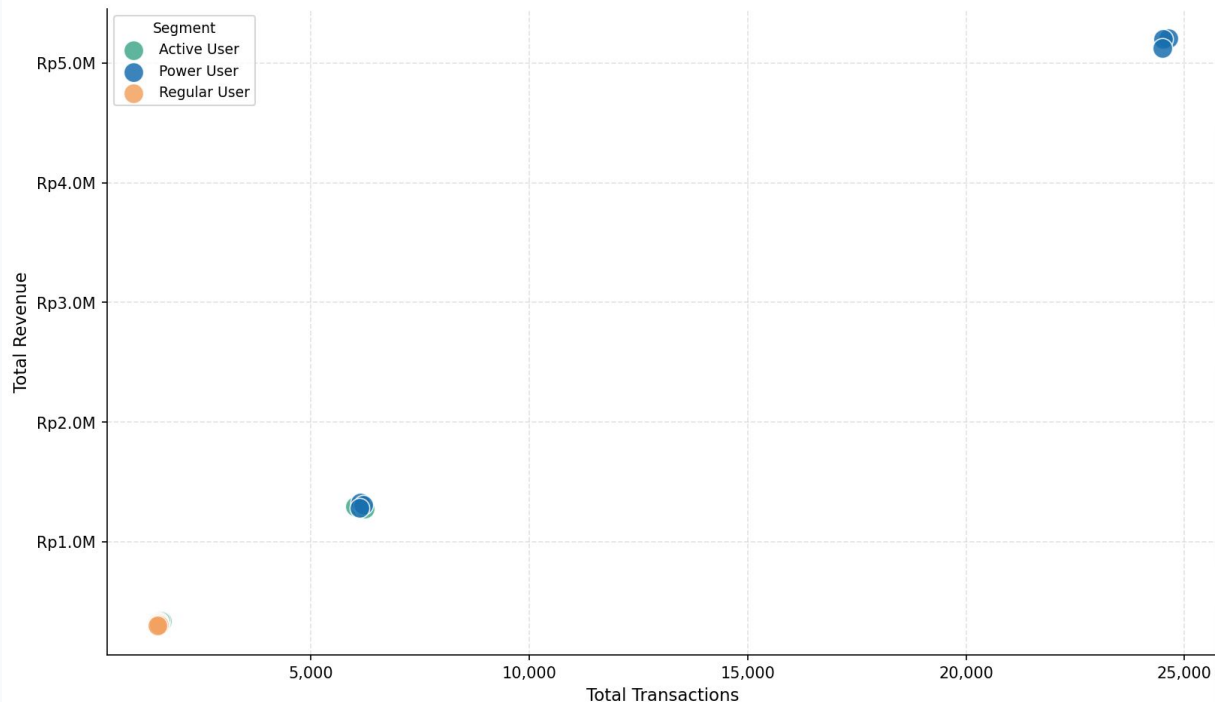
Behavioral Timing

Power Users blast at 08-11 vs Regular at 12-16

~79% of all blasts within business hours across segments

Segmentation: Clustering Overview

User Segmentation: Revenue vs Transaction Volume
(OCA Indonesia — FM-based RFM)



Three distinct tiers visible -- Power Users (3 outliers top-right) generate ~4 the revenue of the Active User cluster, and ~16 Regular Users. No overlap between clusters confirms clean segmentation.

Segmentation Model

FM-based RFM (Rule-based)

Power User

6 clients Rp3.24M avg

FM = 6

Active User

7 clients Rp0.61M avg

FM = 4-5

Regular User

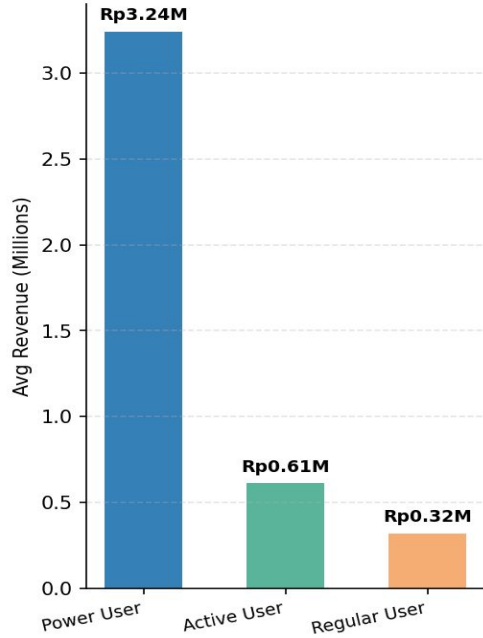
7 clients Rp0.32M avg

FM = 2-3

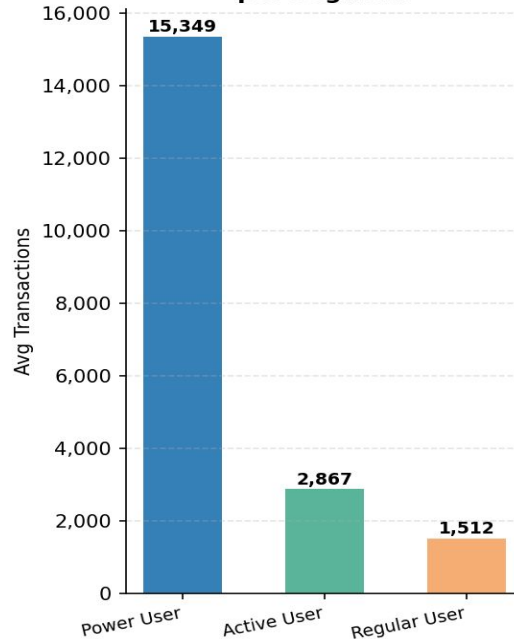
Segmentation: Revenue & Channel Distribution

Segment Summary — OCA Indonesia

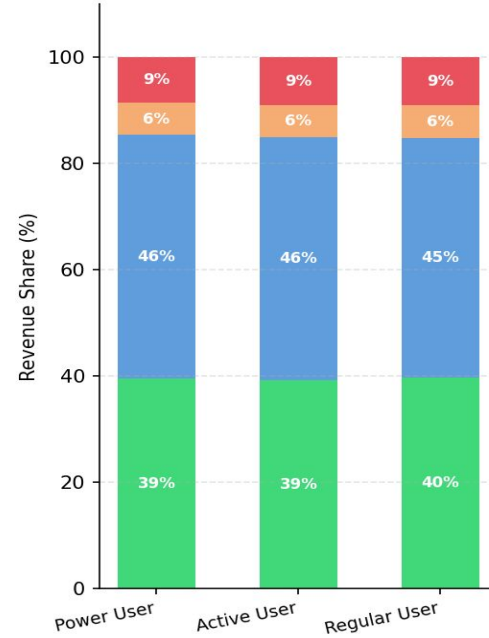
Avg Revenue per Segment



Avg Transaction Volume per Segment



Channel Revenue Mix per Segment

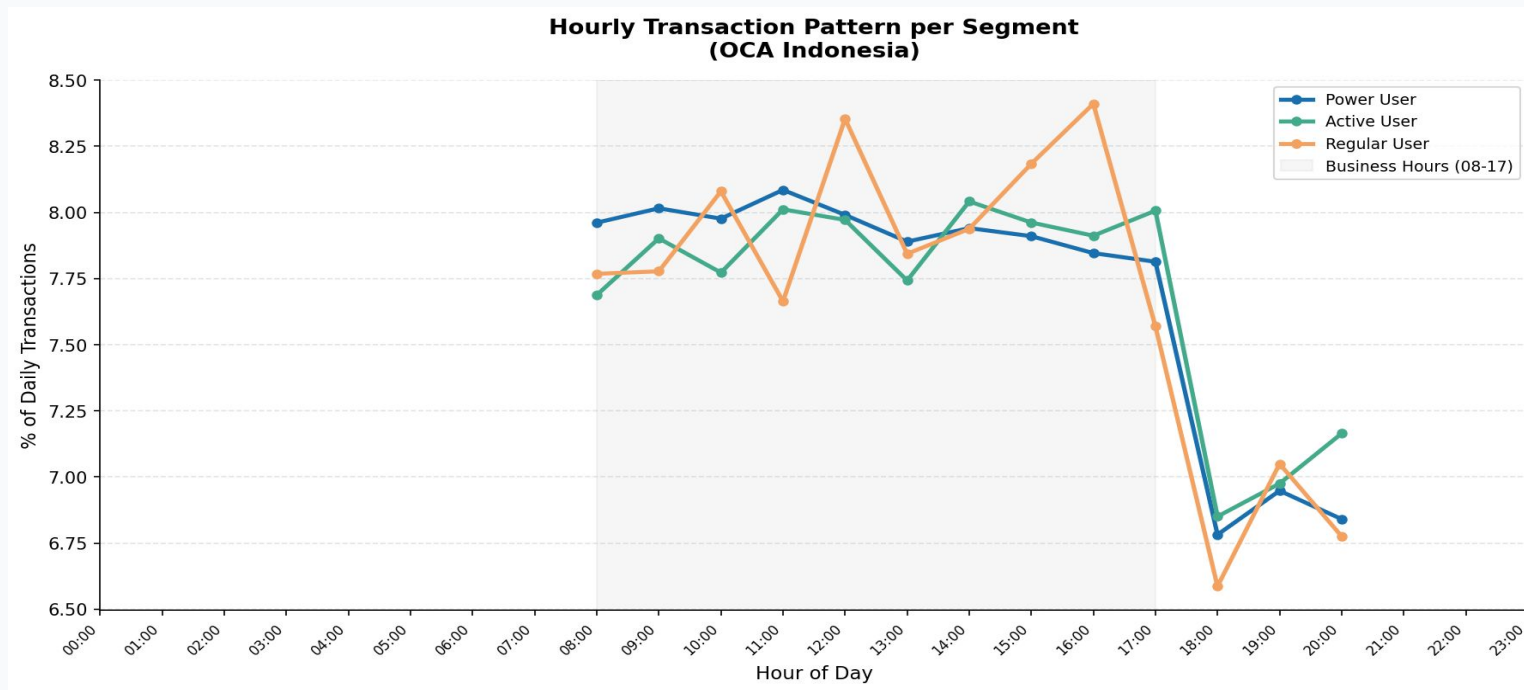


SMS dominates revenue share (~46%) across all three segments. Email and Call contribute modestly (~6% and ~9%). The channel revenue mix is near-identical across segments -- behavior differences lie in volume, not channel preference.

User Persona Table

Segment	# Clients	Revenue Range	Avg Tx / Qtr	Top Channel	Characteristics	Recommended Action
Power User	6	Rp1.28M - Rp5.21M	15,349	SMS + WA (>85%)	Enterprise scale. Daily high-volume blast. All channels active. Morning blasters (08-11).	Loyalty program Dedicated AM Early feature access
Active User	7	Rp1.28M - Rp1.33M	2,867	SMS + WA (~85%)	Mid-size clients. Consistent multi-channel usage. Midday blast pattern (11-15).	Upsell volume package Introduce scheduled blast automation
Regular User	7	Rp0.30M - Rp0.34M	1,512	SMS + WA (~86%)	Smaller clients. Lower volume. Call RNA rate high (76%). Afternoon blast pattern (12-16).	Volume growth incentive Replace Call with WA/SMS to reduce RNA

Insights: Hourly Activity Pattern



Power Users blast early

Peak activity 08-11. Scheduled blasts configured to hit inboxes at business-open.

Active Users peak midday

11:00-15:00 peak. Likely manual or semi-automated campaigns.

Regular Users blast late

12:00-16:00. End-of-day pattern may reduce open rates vs. morning sends.

Insights & Recommendations per Segment

Power User

KEY FINDINGS

- > Generate 65% of total platform revenue with only 30% of clients
- > SMS is dominant revenue channel (~45%); WA second (~39%)
- > Morning blast pattern (08-11) -- highest operational efficiency
- > No churn risk detected -- all active throughout full Q1 period

RECOMMENDATIONS

- ✓ Introduce loyalty / enterprise SLA program
- ✓ Assign dedicated account manager to top 3 users
- ✓ Offer early access to Repeated Blast & API features

Active User

KEY FINDINGS

- > Contribute ~22% of platform revenue -- significant mid-tier
- > Channel mix mirrors Power Users but at 5 lower volume
- > Midday peak (11-15) suggests room for scheduling optimization
- > High SMS sent-rate (99%) indicates operational reliability

RECOMMENDATIONS

- ✓ Upsell to higher volume blast packages
- ✓ Introduce Scheduled Blast with optimal time presets
- ✓ Cross-sell WhatsApp marketing templates

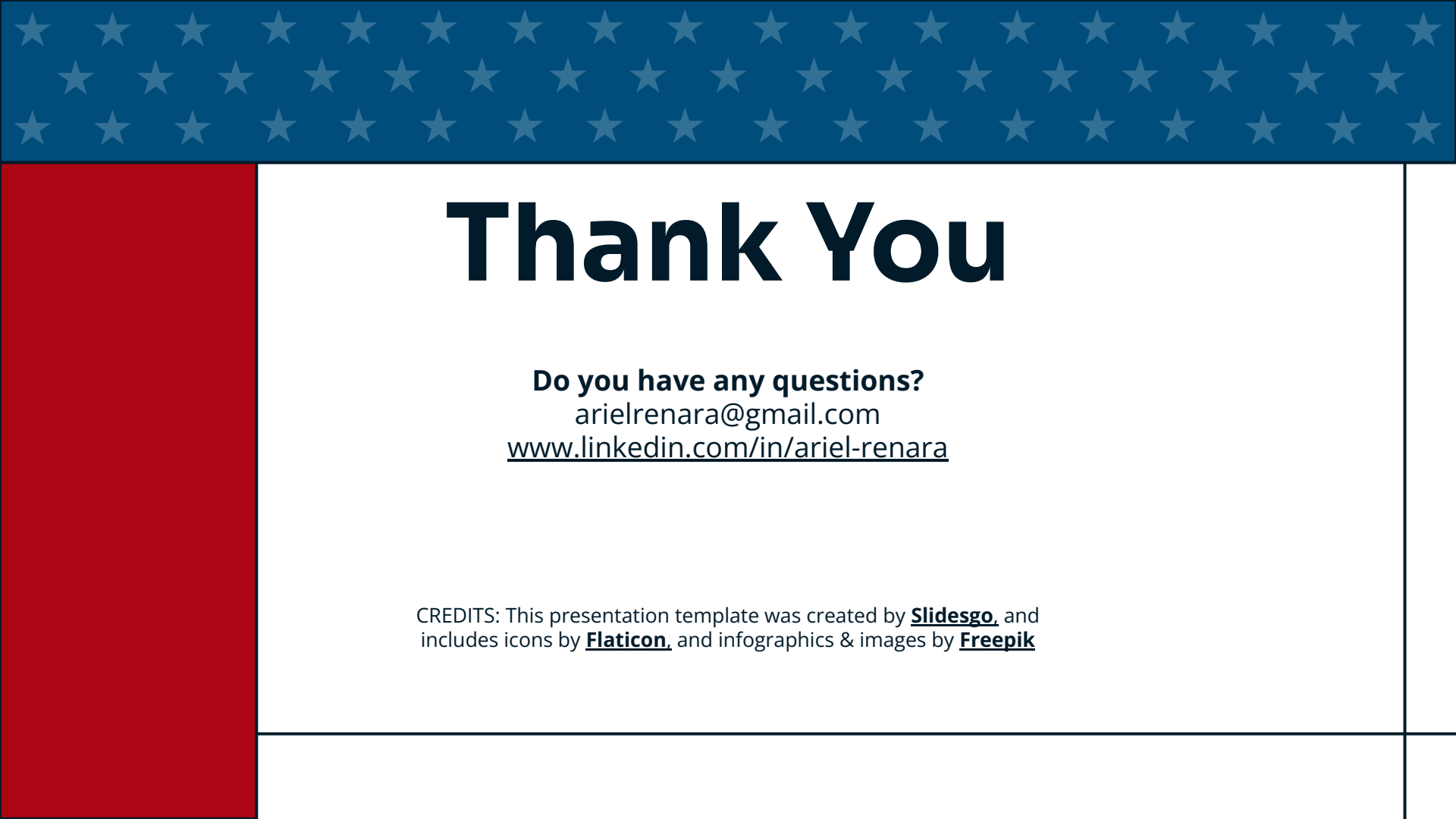
Regular User

KEY FINDINGS

- > 12 clients generating ~13% of revenue -- lowest ARPU tier
- > Call channel shows 76% RNA rate -- high cost, low return
- > Afternoon blast pattern may reduce engagement outcomes
- > Opportunity: shift Call budget to WA/SMS for better ROI

RECOMMENDATIONS

- ✓ Offer volume growth incentive / starter bundle
- ✓ Recommend replacing Call channel with WhatsApp IVR
- ✓ Provide blast scheduling consultation to improve timing



Thank You

Do you have any questions?

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All analyses, insights, and recommendations presented in this deck are based solely on the provided dataset and are intended for academic and portfolio purposes only. Any resemblance to actual business performance, revenue figures, or client identities of OCA Indonesia or Telkom Indonesia is coincidental.

Appendix

Google Colab Python Script

https://colab.research.google.com/drive/1r3F_LlwLkxJqdipPPS7sZ_0Xm6gKZRLO?usp=sharing

Google Bigquery SQL Script

<https://console.cloud.google.com/bigquery?sq=369459387126:a0c2ebd0a1dc48e58965cfab1bb344fe>

Presentation Recording

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